PROJECT OVERVIEW

Customer Order Notification Workflow

Turn spreadsheet orders into clear, consistent team emails.

The workflow

Google Sheets supplies the order data. An n8n workflow passes each order to an OpenAI model, then sends the generated subject and message through Gmail.

Business value

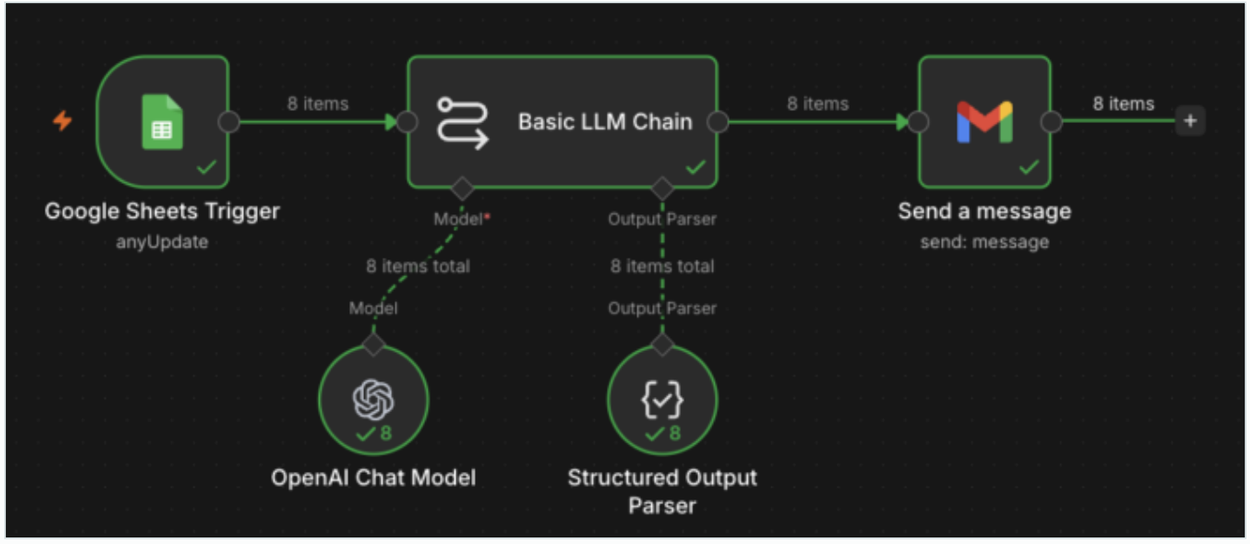
Reduces manual email drafting and keeps order details in a repeatable format for the team.

Tools used

n8n / Google Sheets
OpenAI (GPT-4o) / Gmail

What the supplied run shows
The execution canvas shows eight items passing through the workflow with successful node indicators.

WORKFLOW CANVAS



The Google Sheets Trigger feeds the Basic LLM Chain, which uses an OpenAI Chat Model and Structured Output Parser before Gmail sends the message.

FROM INTAKE TO NOTIFICATION

01

Capture

Read the order row

02

Generate

Create subject + body

03

Deliver

Send through Gmail

STEP 01 / GOOGLE SHEETS TRIGGER

Capture the customer order

Use the "Customer Orders" spreadsheet as the team's order intake. The documented polling interval is every minute.

Order fields

Each row carries the details needed to prepare the team notification.

Order identity	OrderID and CustomerName
Item details	Product and Quantity
Price	Price, preserved as supplied
Order timing	OrderDate
Order state	Status

The sheet stays familiar to non-technical users; no custom intake interface or webhook is required.

> **Trigger configuration**
> The walkthrough describes new rows only. The supplied canvas displays `anyUpdate`; verify the trigger event before describing it as new-row-only.

SOURCE SPREADSHEET / SHEET1

COLUMNS A-C / ORDER IDENTITY AND PRODUCT

OrderID	CustomerName	Product
101	James Harden	Laptop
102	Michael Jordan	Smartphone
103	Stephen Curry	Headphones
104	Klay Thompson	Monitor
105	Shai Alexander	Tablet
106	Test Customer	Keyboard
107	Kobe Bryant	Mouse
108	James Doe	Laptop

The highlighted row is Order #105: Shai Alexander, Tablet.

COLUMNS D-G / QUANTITY, PRICE, DATE AND STATUS

Quantity	Price	OrderDate	Status
1	1200	October 1,2024	Shipped
2	800	October 3,2024	Pending
1	200	October 4,2024	Shipped
1	300	October 5,2024	Delivered
3	600	October 6,2024	Pending
1	75	October 13,2026	Pending
2	599	October 14,2026	Shipped
1	1300	October 15,2026	Pending

The same row supplies quantity 3, price 600, October 6, 2024, and Pending.

STEP 02 / BASIC LLM CHAIN

Generate a structured email

The Basic LLM Chain passes each order to GPT-4o with instructions for writing the team notification.

- **Map the row.** Expressions such as `{{ $json.CustomerName }}` insert the order values into the prompt.
- **Set the format.** Place each order label on its own line and sign the message "Customer Success Team".
- **Return two fields.** The Structured Output Parser validates the expected JSON shape for the Gmail node.

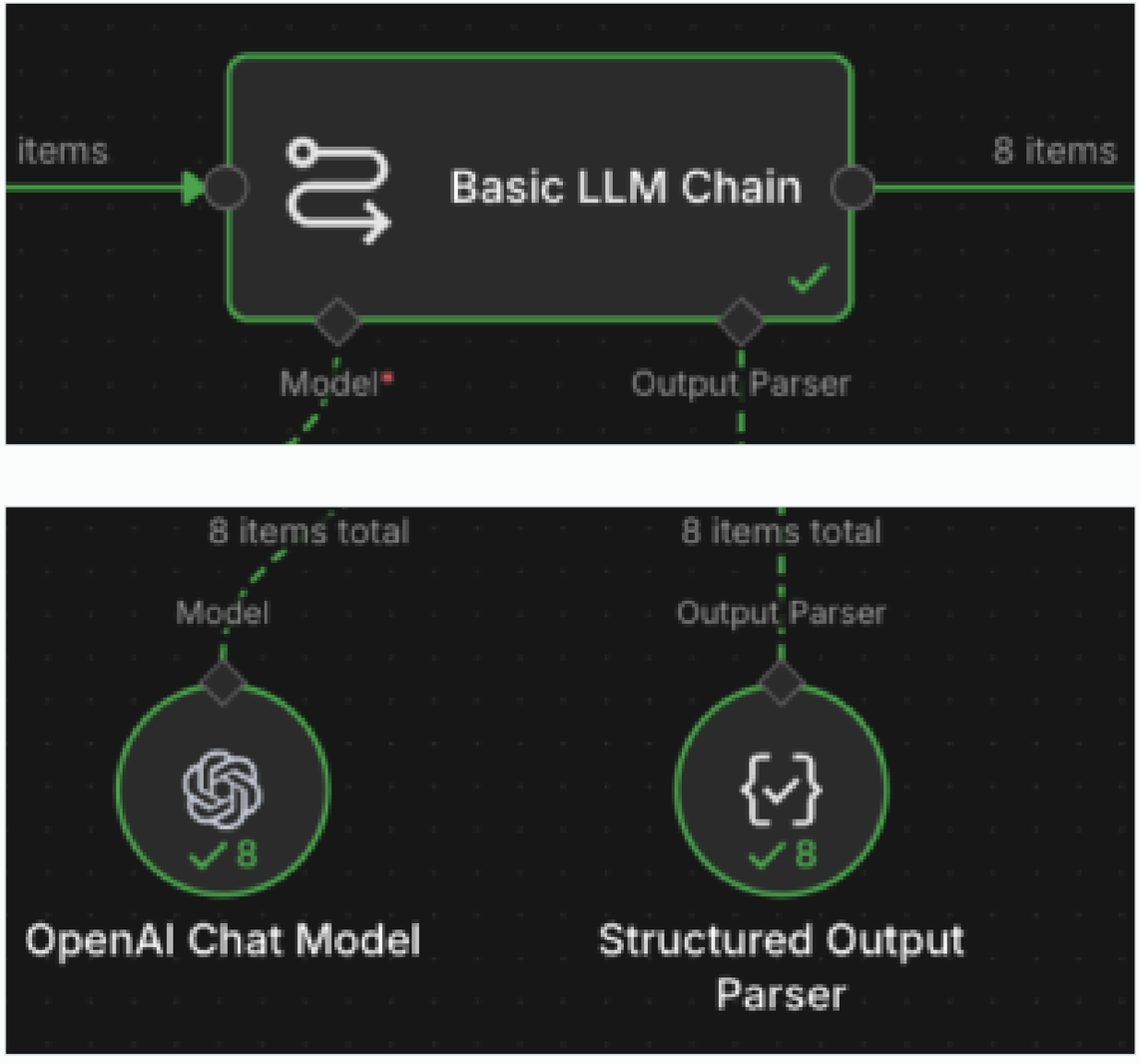
REQUIRED OUTPUT SHAPE

```
{
  "emailSubject": "string",
  "emailBody": "string"
}
```

Both fields are required. Gmail can reference them directly instead of extracting a subject and body from free-form prose.

Schema validation makes the format predictable; it does not guarantee that every model response or execution succeeds.

AI MODEL AND PARSER CONNECTIONS



Two details from the supplied canvas: the chain above and its sub-nodes below. The OpenAI Chat Model connects to Model; the Structured Output Parser connects to Output Parser.

IMPLEMENTATION / DATA FORMATTING

Make dates and line breaks readable

Convert the Sheets date

The walkthrough handles numeric date serials by adding the day count to 1899-12-30, then formatting the result for the email.

DATE EXPRESSION

```
DateTime.fromISO('1899-12-30')
  .plus({
    days: Number($json.OrderDate)
  })
  .toFormat('MMMM d, yyyy')
```

Example: 45930 becomes September 30, 2025.

Keep one detail per line

Gmail receives an HTML body. Replacing newline characters with
 tags preserves the layout requested in the prompt.

```
{{ $json.output.emailBody
  .replace(/\n/g, '<br>') }}
```

The date expression assumes a numeric serial value; the input must not already be a formatted date string.

SOURCE DATA MATCHED TO EMAIL OUTPUT

SOURCE / DATE AND STATUS IN GOOGLE SHEETS

OrderDate	Status
October 1,2024	Shipped
October 3,2024	Pending
October 4,2024	Shipped
October 5,2024	Delivered
October 6,2024	Pending

Order #105 displays October 6, 2024 in the sheet.

OUTPUT / READABLE DATE AND SEPARATE LINES

Order ID: 105
Customer Name: Shai Alexander
Product: Tablet
Quantity: 3
Price: 600
Order Date: October 6, 2024
Status: Pending

The email preserves the supplied order details and displays the date as October 6, 2024.

STEP 03 / GMAIL - SEND A MESSAGE

Deliver the team notification

The Gmail node sends the AI-generated subject and body to the configured recipient.

SUBJECT MAPPING

```
{{ $json.output.emailSubject }}
```

Body and presentation

Use the generated emailBody with the newline replacement shown on the previous page. The walkthrough disables the "sent via n8n" attribution footer.

Recipient

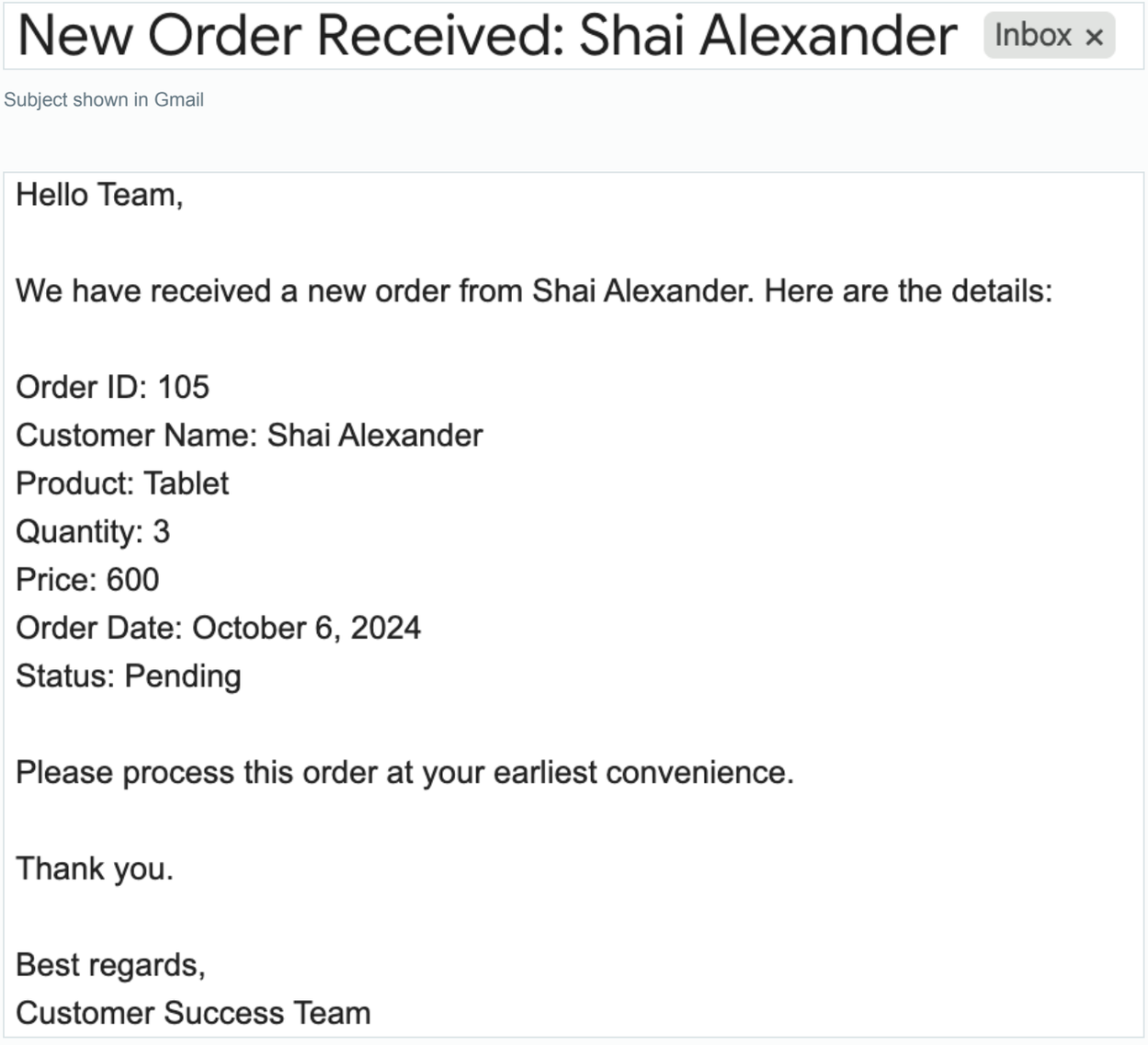
The documented setup uses one address. A team distribution list can be used for a shared operations inbox.

Example shown

Order #105 / Shai Alexander
Tablet / Quantity 3 / Price 600
October 6, 2024 / Pending

Sample result: a team notification for Order #105 using the values shown in the source spreadsheet. The price is preserved without adding an assumed currency.

DELIVERED EMAIL / SUPPLIED SCREENSHOT



Subject shown in Gmail

Sender details are removed from the portfolio copy. Email content is retained.

DESIGN DECISIONS / NEXT STEPS

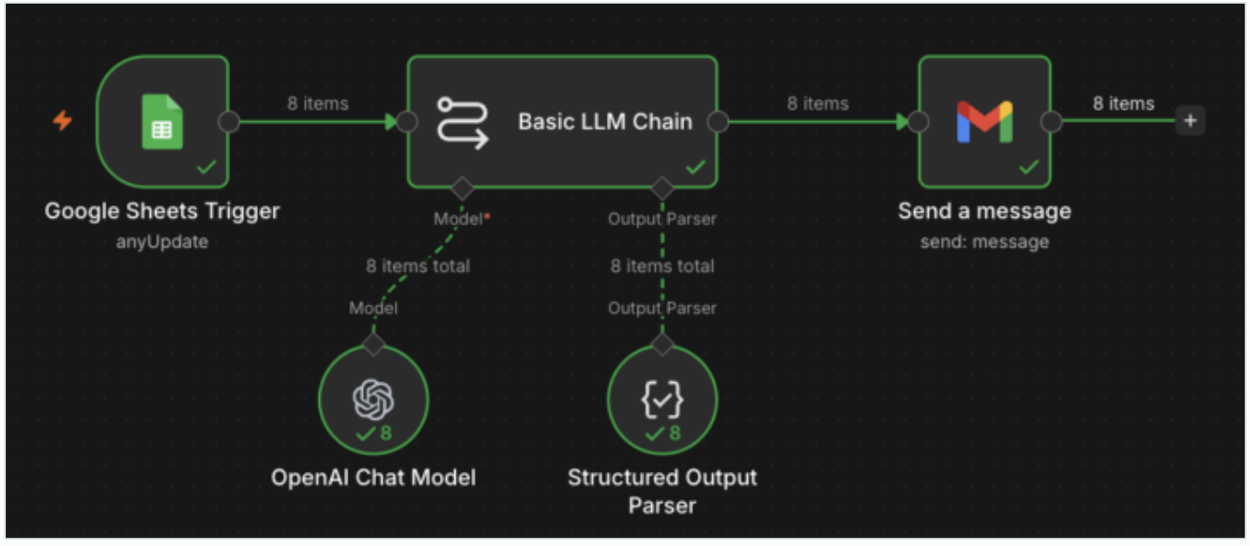
Why this design works

- **Google Sheets.** A familiar place for the team to enter orders.
- **Structured output.** A defined subject and body for the Gmail handoff.
- **Prompt layout.** Consistent labels and a team signature.
- **Date + HTML formatting.** Readable dates and separate lines in the email.

POSSIBLE IMPROVEMENTS

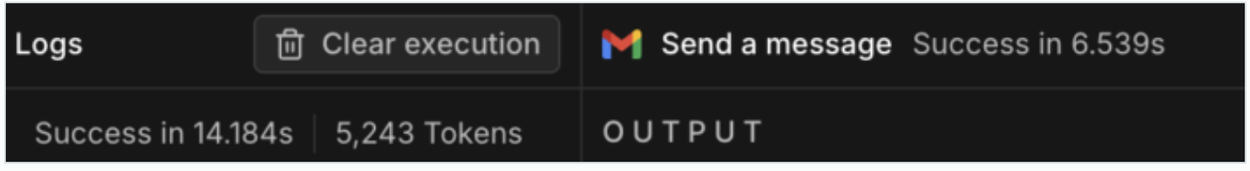
- **Error alerts.** Notify an owner if the model or Gmail step fails.
- **Notification tracking.** Write back a Notified status and check it before sending to reduce repeat notifications.
- **Routing.** Send by product or order size; optionally add Slack or Teams.
- **Status filter.** Process only eligible rows, such as those marked New.

EXECUTION EVIDENCE



Green node indicators show a successful execution in the supplied canvas.

RECORDED EXECUTION



The screenshot records Success in 14.184s for the run and Success in 6.539s for Gmail. These are observed results from one execution.

Evidence included: the input spreadsheet, connected n8n workflow, and a received order notification in Gmail.